

HYDROGEN PRODUCTION

Engineering Learning Modules

From basics to industrial plant engineering

10 modules · diagrams · comparison tables · calculator · quizzes

Flashcards · assessment · glossary · progress tracker

Prepared for: Ashishpreet · Mechanical / manufacturing engineering background

Edition: October 2026 · Sources: IRENA, IEA, US DOE, ISO/IEC and peer-reviewed references (Appendix E)

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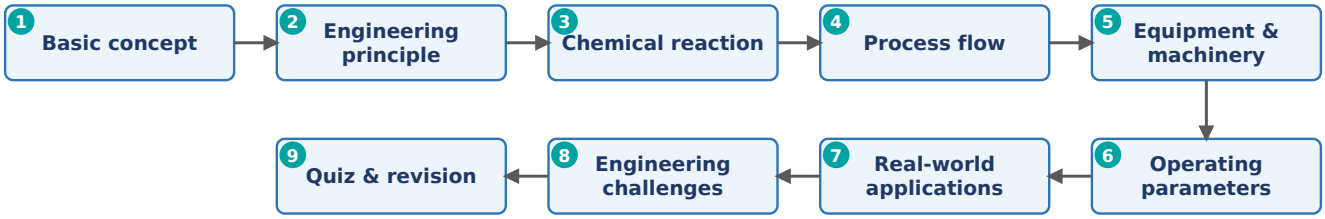
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How to use these modules

Each topic follows the same **nine-step learning sequence**, from simple to advanced:



Symbol / colour	Meaning
Green maturity bar	Commercially established technology
Orange maturity bar	Emerging: demonstration / early commercial
Red maturity bar	Laboratory / pilot stage
Mint box	Chemical reaction or equation
Amber box	Simplified estimate or safety scenario – not plant data
[R1] … [R10]	Reference in Appendix E

About accuracy
Numbers come from the referenced sources. Where a reliable public value was not found (for example, licensor-specific ATR conditions), the parameter is described in words rather than invented. Module 6 values marked [assumption] are teaching inputs only. Electrolyser data are the IRENA 2020 benchmark – always confirm with current OEM data sheets.

Suggested study plan

- Week 1: Modules 1–2 · Week 2: Modules 3–4 (core) · Week 3: Modules 5–7 · Week 4: Modules 8–10 + Appendix B assessment.
- After each module: do the quiz, fill the Progress Tracker (Appendix D), revise flashcards for any wrong answer.

MODULE 1

Introduction to Hydrogen

Properties · energy carrier · colours · demand

Learning objectives

- State the key physical and combustion properties of hydrogen and what they mean for engineering design.
- Explain why hydrogen is an **energy carrier**, not a primary energy source.
- Compare hydrogen with electricity, natural gas and diesel on energy per kg and per litre.
- Classify hydrogen by production route (colour codes) and by emissions intensity.
- Describe where hydrogen is used today and where new demand is expected.

1.1 Basic concept

Hydrogen (H_2) is the lightest molecule. It does not occur free in useful quantities on Earth, so it must be **produced** from a hydrogen-containing compound – mainly water (H_2O) or hydrocarbons such as methane (CH_4) and coal. Energy is spent to make it, and a part of that energy is recovered when hydrogen is burned or used in a fuel cell. That is why hydrogen is described as an **energy carrier** – like electricity – rather than an energy source.

1.2 Engineering properties you will use

Property	Value	Engineering meaning
Molar mass	2.016 g/mol	Very light; leaks easily through small gaps and permeates some materials
Gas density (0 °C, 1 atm)	0.0899 kg/m³ (≈1/14 of air)	Rises and disperses quickly outdoors; accumulates under roofs indoors
Normal boiling point	–252.9 °C (20.3 K)	Liquid H_2 needs cryogenic, vacuum-insulated equipment
Liquid density	≈71 kg/m³	Even liquid H_2 is ~10× lighter than diesel – volume is the challenge
Lower heating value (LHV)	120 MJ/kg = 33.3 kWh/kg	Basis for fuel-cell / engine efficiency
Higher heating value (HHV)	142 MJ/kg = 39.4 kWh/kg	Basis for electrolyser efficiency (water leaves as liquid)
Flammability range in air	4 – 75 vol %	Very wide range – leaks must be detected and ventilated
Minimum ignition energy	≈0.017 mJ (in air)	Very easy to ignite – control static and ignition sources
Auto-ignition temperature	≈585 °C (858 K)	Gas group IIC, temperature class T1 in hazardous-area design

Values: Stolten (ed.), *Hydrogen and Fuel Cells*, Wiley-VCH, ch. 1 [R6]; LHV/HHV kWh conversion used by IRENA [R1].

1.3 Hydrogen versus other energy carriers

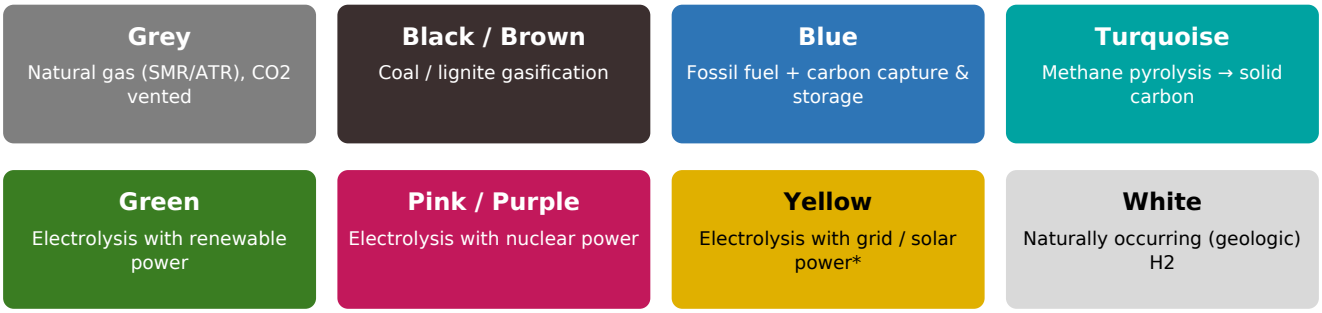
Carrier	Energy per kg	Energy per volume	Key point
Hydrogen	Highest (33.3 kWh/kg LHV)	Lowest – needs compression, liquefaction or a carrier molecule	Excellent per kg, poor per litre
Natural gas (CH_4)	Lower than H_2	≈3× higher than H_2 gas at the same pressure	Existing pipelines and know-how

Carrier	Energy per kg	Energy per volume	Key point
Diesel / petrol	Lower than H_2	Very high – liquid at room temperature	Easy storage; CO_2 on combustion
Electricity (battery)	Low per kg of battery	Medium	Highest round-trip efficiency; no molecule to store

Engineering principle – why conversion losses matter

Every conversion step loses energy: electricity → electrolysis → compression or liquefaction → transport → fuel cell → electricity. Where a direct electric solution exists (e.g. a battery car on short duty), it usually wins on efficiency. Hydrogen is most valuable where you need a **molecule** (ammonia, refining, steel reduction), **long-duration storage**, or **high energy per kg** (heavy, long-haul transport).

1.4 Hydrogen classifications (colour codes)



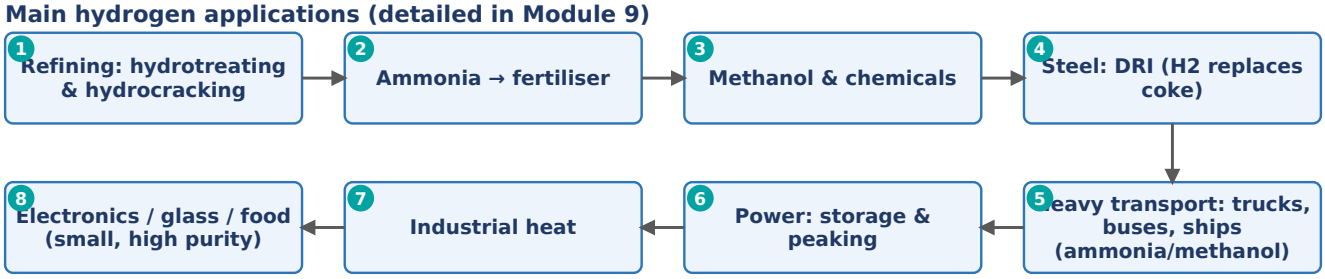
*Colour names are informal and vary by country (*yellow is used for both solar and grid electricity). The IEA recommends classifying hydrogen by **emissions intensity** (kg CO_2-eq per kg H_2) instead [R5].*

Production route	Emissions intensity (kg CO_2-eq / kg H_2)	Source
Unabated natural gas (SMR)	10 – 14 (of which 1 – 5 upstream/midstream)	IEA [R5]
Natural gas + CCS, ~60 % capture	≈5 – 8	IEA [R5]
Natural gas + CCS, >90 % capture	0.8 – 6	IEA [R5]
Unabated fossil (worst cases, incl. coal)	up to ≈27	IEA [R5]
Electrolysis with renewable electricity	0 (IPHE method, production stage)	IEA [R5]
Global average production (2021)	12 – 13	IEA [R5]
India Green Hydrogen Standard threshold	≤ 2 (well-to-gate, 12-month average)	MNRE, Aug 2023 [R10]

1.5 Demand and applications

Global hydrogen demand was **above 97 Mt in 2023** [R3] and **almost 100 Mt in 2024**, up 2 % [R4]. Demand is concentrated in **oil refining** and **industry** (ammonia, methanol, steel via direct reduction). New uses – heavy industry, long-distance transport, energy storage – were still **below 1 %** of demand in 2023, and low-emissions hydrogen production was under 1 Mt [R3]. China is the largest producer of fuel-cell commercial vehicles (≈95 % of world stock) [R4].

Main hydrogen applications (detailed in Module 9)



1.6 Engineering challenges

- **Cost** – low-emissions hydrogen is still more expensive than unabated fossil hydrogen; the gap is expected to narrow [R4].
- **Storage density** – very low energy per litre at ambient conditions.
- **Materials** – embrittlement of high-strength steels; permeation and leakage.
- **Safety** – wide flammability range and very low ignition energy; invisible flame in daylight.
- **Infrastructure** – pipelines, refuelling stations and certification schemes are still being built.

Quiz & revision – Module 1

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Why is hydrogen called an energy carrier?
a) It is found free in nature **b)** It must be produced using another energy source **c)** It is a fossil fuel **d)** It cannot be burned

Q2. Which heating value is used for electrolyser efficiency?
a) LHV 33.3 kWh/kg **b)** HHV 39.4 kWh/kg **c)** Either – they are equal **d)** Neither

Q3. Flammability range of H2 in air:
a) 5–15 % **b)** 1–8 % **c)** 4–75 % **d)** 20–40 %

Q4. IEA recommends classifying hydrogen by:
a) Colour **b)** Price **c)** Emissions intensity **d)** Purity

Q5. Largest current use of hydrogen:
a) Passenger cars **b)** Refining & industry **c)** Aviation **d)** Home heating

Answer key
Q1 – b: it must be made from water or hydrocarbons; **Q2 – b**: water leaves as liquid in the reference reaction; **Q3 – c**: 4–75 vol %; **Q4 – c**: kg CO2-eq per kg H2; **Q5 – b**: refining, ammonia, methanol, steel

MODULE 2

Hydrogen Production Technologies

Nine routes – principle, reactions, equipment, maturity

How to read this module
Each technology follows the same card: **principle → reactions → flow → equipment → conditions → energy → purity → pros/cons → applications**. The coloured bar shows maturity: **green = commercial**, **orange = demonstration / early commercial**, **red = laboratory / pilot**. Where a reliable public figure was not found, the parameter is described qualitatively rather than invented.

2.1 Steam Methane Reforming (SMR)

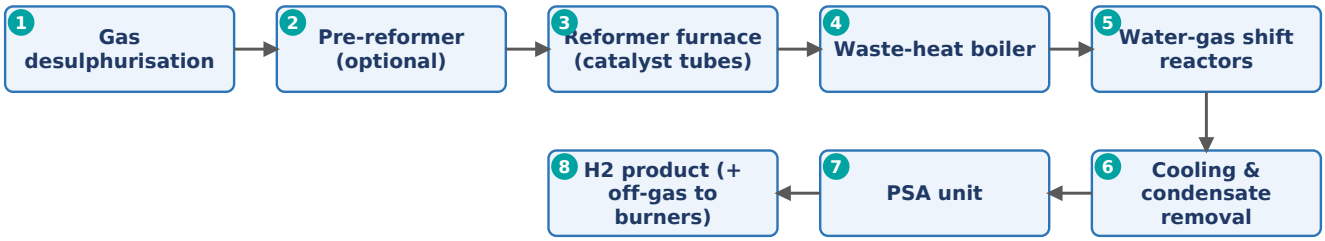
Technology maturity: Commercial

Working principle. Methane reacts with high-temperature steam over a nickel-based catalyst in tubes inside a fired furnace. The reaction is endothermic, so heat is supplied by burning fuel (natural gas plus PSA off-gas). A water-gas shift step then converts CO into more H_2, and pressure swing adsorption purifies the product [R2].

$$CH_4 + H_2O + \text{heat} \rightarrow CO + 3H_2$$
Reforming (endothermic, ΔH ≈ +206 kJ/mol)

$$CO + H_2O \rightarrow CO_2 + H_2 + \text{small heat}$$
Water-gas shift (exothermic, ΔH ≈ –41 kJ/mol)

Process flow



Item	Details
Raw materials	Natural gas (or LPG, naphtha, biogas), process steam (boiler feed water), combustion air
Main equipment	Desulphuriser (ZnO beds), reformer furnace with catalyst-filled tubes and burners, waste-heat/steam boiler, HT/LT shift reactors, process gas coolers, condensate separators, PSA adsorber beds, product compressor; for blue H_2: CO_2 capture unit
Operating conditions	Reformer: **700 – 1,000 °C** at **3 – 25 bar** with catalyst [R2]. Shift: lower temperature, catalytic.
Energy requirement	Thermal energy (furnace firing) dominates; natural gas is both feedstock and fuel.
Hydrogen purity	PSA delivers essentially pure hydrogen [R2] (the CO_2 and impurities are removed).
Advantages	Mature, large single-train capacity, lowest cost where gas is cheap, huge operating experience
Disadvantages	CO_2 emissions 10–14 kg CO_2-eq/kg H_2 unabated [R5]; flue-gas CO_2 is dilute and costly to capture
Industrial applications	Refineries, ammonia and methanol plants, merchant hydrogen

2.2 Autothermal Reforming (ATR)

Technology maturity: Commercial

Working principle. ATR combines partial oxidation and steam reforming inside one refractory-lined reactor. Oxygen and steam are fed with the gas; the exothermic oxidation supplies the heat for the endothermic reforming, so no external furnace is needed. Because there is no fired furnace, almost all CO_2 ends up in the high-pressure process stream – which makes carbon capture easier. ATR is therefore the preferred route for many large **blue hydrogen** projects.

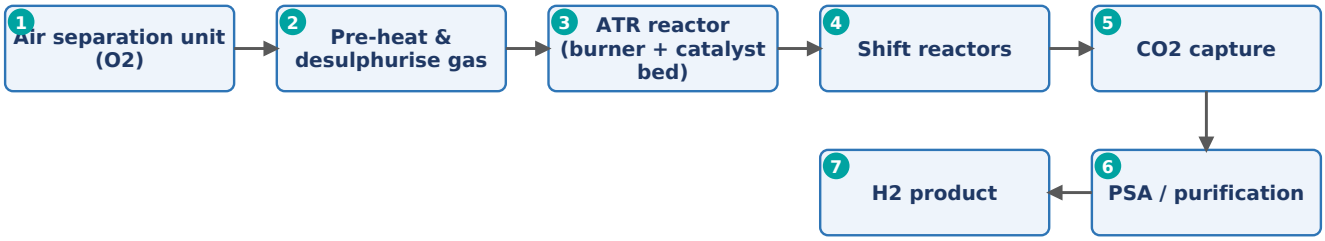
$$CH_4 + ½O_2 \rightarrow CO + 2H_2 \text{ (+ heat)}$$

Partial oxidation zone

$$CH_4 + H_2O \rightarrow CO + 3H_2 \text{ (uses that heat)}$$

Catalytic reforming zone, then water-gas shift

Process flow



Item	Details
Raw materials	Natural gas, oxygen (from an air separation unit), steam
Main equipment	Air separation unit, ATR reactor with burner and catalyst bed, waste-heat boiler, shift reactors, amine (or similar) CO_2 capture, PSA
Operating conditions	High-temperature, oxygen-blown; specific temperatures and pressures are licensor-specific (not given here to avoid unsourced values).
Energy requirement	Self-heated reactor; electricity for the air separation unit is a significant load.
Hydrogen purity	High after shift + PSA / purification
Advantages	No furnace; concentrated CO_2 stream; large single-train size; well suited to CCS
Disadvantages	Needs an oxygen plant (capital + power); still fossil-based; CCS needs storage site
Industrial applications	Large ammonia/methanol plants; blue hydrogen hubs

2.3 Partial Oxidation (POX)

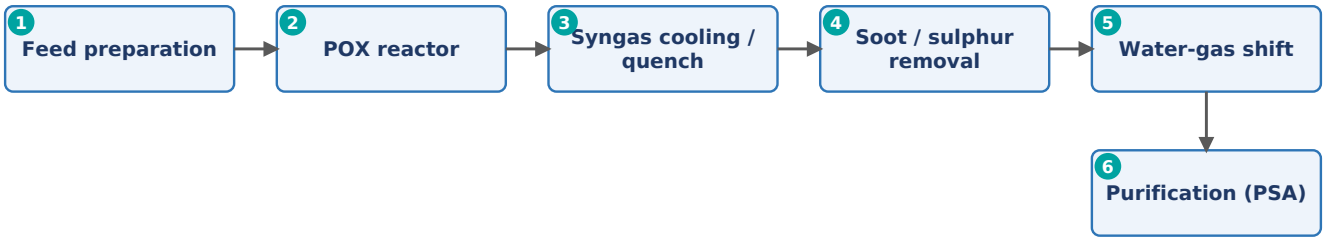
Technology maturity: Commercial

Working principle. Methane or heavier hydrocarbons react with a limited amount of oxygen. The reaction is exothermic, typically faster than steam reforming and needs a smaller reactor, but gives less hydrogen per unit of fuel [R2]. Shift and purification follow as in SMR.

$$CH_4 + ½O_2 \rightarrow CO + 2H_2 + \text{heat}$$

Exothermic; air as oxidant adds N2 to the product [R2]

Process flow



Item	Details
Raw materials	Natural gas, heavy oil residues, petroleum coke; oxygen or air
Main equipment	Oxygen plant, refractory POX gasifier, quench or syngas cooler, soot scrubber, acid-gas removal, shift reactors, PSA
Operating conditions	Non-catalytic, high-temperature operation; conditions depend on feedstock and licensor.
Energy requirement	Self-heated; oxygen production is the main parasitic energy load.
Hydrogen purity	Syngas contains CO, CO_2, some N_2 (if air-blown) – purified by shift + PSA
Advantages	Handles heavy and dirty feedstocks; compact, fast reaction
Disadvantages	Lower H_2 yield than SMR; oxygen plant cost; CO_2 emissions
Industrial applications	Refinery residue upgrading, syngas for chemicals

2.4 Water Electrolysis (overview)

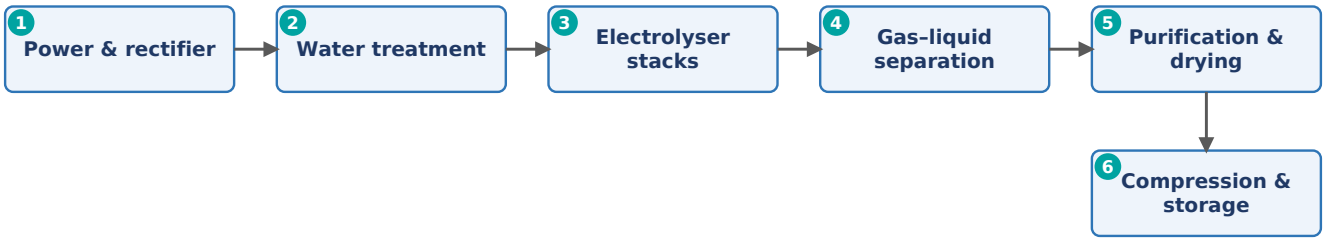
Technology maturity: Commercial

Working principle. Direct current splits water into hydrogen at the cathode and oxygen at the anode. With renewable or nuclear electricity the process emits no CO_2 at the production stage. Four cell types exist – alkaline (AEL), proton exchange membrane (PEM), solid oxide (SOEC) and anion exchange membrane (AEM) – detailed in Module 4.

$$2H_2O + \text{electricity} \rightarrow 2H_2 + O_2$$

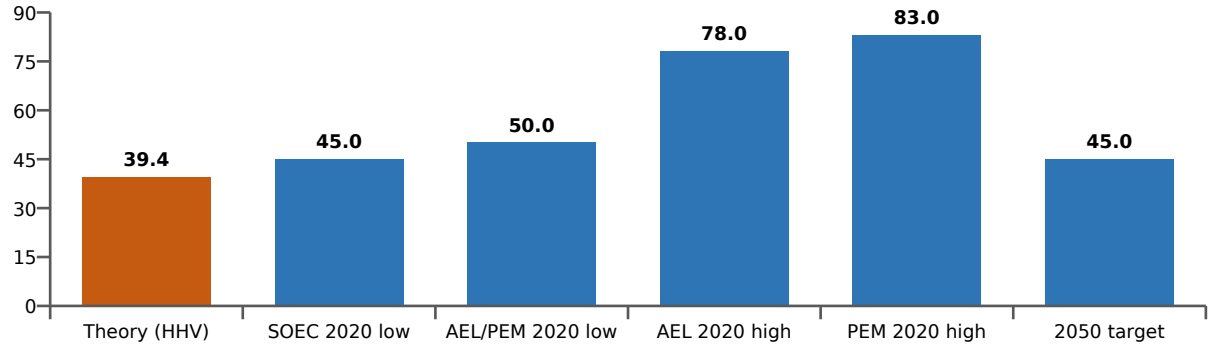
ΔH = 285.8 kJ/mol H2 (HHV) · reversible voltage 1.23 V · thermoneutral voltage 1.48 V (25 °C)

Process flow



Item	Details
Raw materials	Demineralised water (9 kg per kg H_2 stoichiometric; 18 – 24 kg incl. demineralisation losses [R1]); DC electricity
Main equipment	Transformer–rectifier, stacks, separators, circulation pumps, heat exchangers, de-oxo, dryer, compressor
Operating conditions	AEL 70–90 °C / 1–30 bar; PEM 50–80 °C / <70 bar; AEM 40–60 °C / <35 bar; SOEC 700–850 °C / ~1 bar [R1]
Energy requirement	System 2020: AEL 50–78, PEM 50–83, AEM 57–69, SOEC 45–55 kWh/kg H_2; 2050 target <45 (SOEC <40) [R1]
Hydrogen purity	Very high after de-oxo and drying – suitable for fuel cells
Advantages	Zero direct emissions with clean power; modular; flexible operation (esp. PEM); O_2 by-product
Disadvantages	Electricity cost dominates; stack capital cost; critical materials (Ir, Pt for PEM)
Industrial applications	Green ammonia, refineries, steel DRI, refuelling stations, energy storage

Electricity per kg H2 (kWh/kg) – theory vs. system (IRENA, 2020)



Theoretical minimum = HHV 39.4 kWh/kg. Real systems need more because of overpotentials, ohmic losses and balance-of-plant loads [R1].

2.5 Biomass Gasification

Technology maturity: Early commercial / demonstration

Working principle. Biomass is heated with a controlled amount of oxygen and/or steam (without full combustion) to make syngas (H_2, CO, CO_2, CH_4). Tars and particulates are removed, CO is shifted to H_2, and the gas is purified.

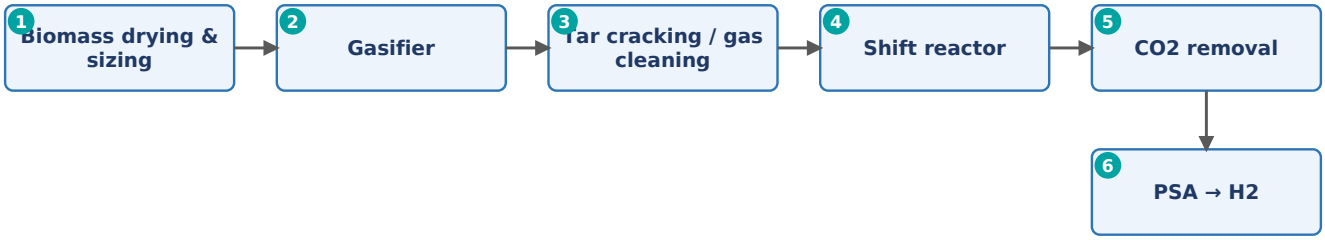
$$\text{Biomass} + O_2 / H_2O \rightarrow CO + H_2 + CO_2 + CH_4 + \text{tar}$$

Gasification

$$CO + H_2O \rightarrow CO_2 + H_2$$

Water-gas shift

Process flow



Item	Details
Raw materials	Agricultural residues, wood chips, energy crops, municipal solid waste (bio-fraction)
Main equipment	Feed handling, dryer, gasifier (fixed bed / fluidised bed / entrained flow), cyclones, tar reformer, scrubbers, shift, PSA
Operating conditions	Gasifier temperature and pressure depend on design; tar control is the key operating issue.
Energy requirement	Thermal energy from partial oxidation of the biomass itself
Hydrogen purity	High after PSA
Advantages	Uses waste/residue; can be carbon-neutral or negative with CCS
Disadvantages	Variable feed quality; tars foul equipment; logistics of collecting biomass; smaller scale
Industrial applications	Distributed hydrogen near agri-residue sources; waste-to-hydrogen

2.6 Coal Gasification

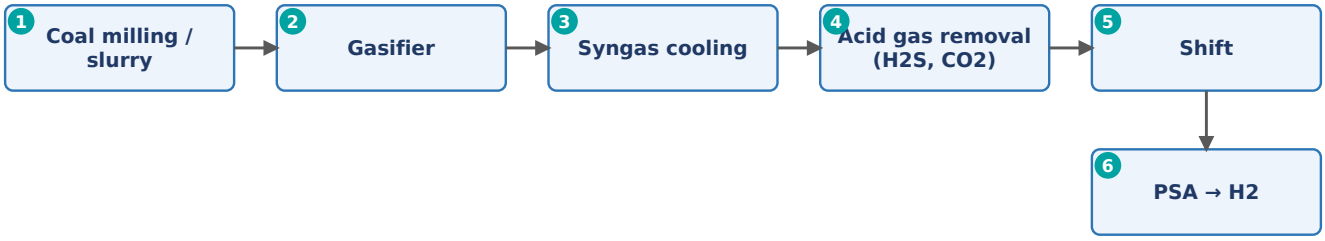
Technology maturity: Commercial

Working principle. Coal is gasified with oxygen and steam in an entrained-flow or fluidised-bed gasifier to syngas, which is cleaned, shifted and purified. Widely used in China for ammonia and methanol.

$$C + H_2O \rightarrow CO + H_2$$
Steam gasification (endothermic)

$$C + \tfrac{1}{2}O_2 \rightarrow CO$$
Partial combustion supplies heat

Process flow



Item	Details
Raw materials	Coal or lignite, oxygen, steam
Main equipment	Air separation unit, gasifier, syngas coolers, acid-gas removal, sulphur recovery, shift reactors, PSA
Operating conditions	High-temperature, oxygen-blown, pressurised gasifiers (design-specific)
Energy requirement	Large; plus oxygen production
Hydrogen purity	High after purification
Advantages	Uses abundant domestic coal; mature large plants
Disadvantages	Highest CO_2 intensity of all routes (up to ≈27 kg CO_2-eq/kg incl. upstream [R5]); ash and water handling
Industrial applications	Ammonia, methanol, coal-to-chemicals

2.7 Methane Pyrolysis (turquoise hydrogen)

Technology maturity: Early commercial / demonstration

Working principle. Methane is cracked at high temperature without oxygen into hydrogen and **solid carbon**. No CO_2 is formed in the reaction; the carbon (carbon black, graphite) can be sold. Heat comes from plasma, molten metal/salt baths or catalysts.

$$CH_4 + \text{heat} \rightarrow C(s) + 2H_2$$

ΔH ≈ +75 kJ/mol CH4 – much less energy per kg H2 than electrolysis

Process flow



Item	Details
Raw materials	Natural gas or biomethane; electricity or heat
Main equipment	Plasma torch reactor or molten-metal bubble column, carbon separation (cyclones, filters), gas clean-up / PSA
Operating conditions	Very high temperature; design-specific
Energy requirement	Lower electricity per kg H_2 than electrolysis (theoretical)
Hydrogen purity	High after clean-up
Advantages	No process CO_2; saleable carbon by-product
Disadvantages	About 40 % of the methane's heating value stays in the solid carbon (802 vs 2×242 kJ/mol LHV); carbon market size; reactor fouling; upstream methane leaks still count
Industrial applications	Carbon black + hydrogen co-production

2.8 Photoelectrochemical (PEC) Water Splitting

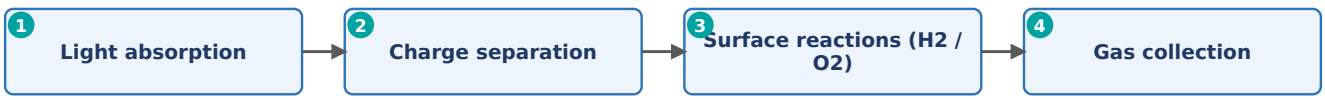
Technology maturity: Laboratory

Working principle. A semiconductor electrode immersed in water absorbs sunlight and directly drives water splitting – combining a solar cell and an electrolyser in one device.

$$2H_2O + \text{sunlight} \rightarrow 2H_2 + O_2$$

Photo-generated electron–hole pairs drive the half-reactions

Process flow



Item	Details
Raw materials	Water, sunlight, semiconductor photoelectrodes
Main equipment	Photo-electrodes (metal oxides, III-V semiconductors), membranes, transparent reactor panels
Operating conditions	Ambient temperature, sunlight
Energy requirement	Solar energy only
Hydrogen purity	Requires drying/purification
Advantages	Direct solar-to-hydrogen; potentially simple
Disadvantages	Low efficiency and durability (corrosion) today; not yet scaled

Item	Details
Industrial applications	Research

2.9 Biological Hydrogen Production

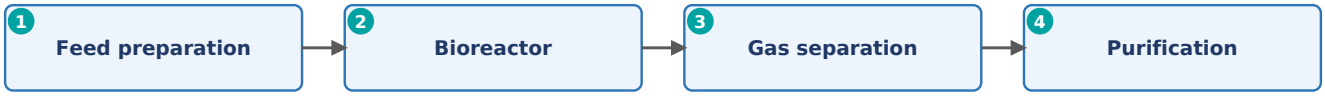
Technology maturity: Laboratory / pilot

Working principle. Micro-organisms produce hydrogen: algae and cyanobacteria by **photobiological** water splitting, and bacteria by **dark fermentation** of sugars/organic waste or photo-fermentation.

$$C_6H_{12}O_6 + 2H_2O \rightarrow 2CH_3COOH + 2CO_2 + 4H_2$$

Dark fermentation (acetate pathway)

Process flow



Item	Details
Raw materials	Organic waste, wastewater, sugars; sunlight for photobiological routes
Main equipment	Bioreactors / photobioreactors, gas separation membranes, PSA
Operating conditions	Near-ambient temperature
Energy requirement	Low external energy
Hydrogen purity	Gas contains CO_2; must be purified
Advantages	Uses wastes at low temperature
Disadvantages	Low production rates and yields; large reactor volumes
Industrial applications	Research; possible niche in waste treatment

2.10 Summary comparison

Route	Feedstock	Direct CO_2	Maturity	Main limitation
SMR	Natural gas + steam	Yes (dilute)	Commercial	Emissions
ATR	Gas + O2 + steam	Yes (concentrated)	Commercial	Needs CCS + O2 plant
POX	Gas / heavy oil + O2	Yes	Commercial	Lower yield
Electrolysis AEL/PEM	Water + power	No	Commercial	Power cost
Electrolysis SOEC/AEM	Steam/water + power	No	Lab-scale (IRENA 2020) → scaling	Durability, scale
Biomass gasification	Biomass	Biogenic	Demonstration	Tars, logistics
Coal gasification	Coal	Yes (highest)	Commercial	Emissions
Methane pyrolysis	Gas	No (solid C)	Early commercial	Carbon market, scale
PEC	Water + sun	No	Laboratory	Efficiency
Biological	Waste / sun	Biogenic	Laboratory / pilot	Rate

Quiz & revision – Module 2

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. SMR reformer conditions per US DOE:
a) 200–300 °C, 100 bar **b)** 700–1,000 °C, 3–25 bar **c)** 1,500 °C, 1 bar **d)** 80 °C, 30 bar

Q2. Why is ATR preferred for blue hydrogen?
a) Uses no oxygen **b)** CO2 is concentrated in the process stream **c)** Has no catalyst **d)** Produces no CO

Q3. By-product of methane pyrolysis:
a) CO2 **b)** Solid carbon **c)** Ammonia **d)** Oxygen

Q4. Stoichiometric water per kg H2:
a) 1 kg **b)** 4.5 kg **c)** 9 kg **d)** 24 kg

Q5. Which route is still laboratory-stage?
a) SMR **b)** Coal gasification **c)** PEC water splitting **d)** PEM electrolysis

Answer key
Q1 – b: DOE [R2]; **Q2 – b**: no fired furnace, capture is easier; **Q3 – b**: carbon black/graphite; **Q4 – c**: 18 g water → 2 g H2; **Q5 – c**: photo-electrochemical

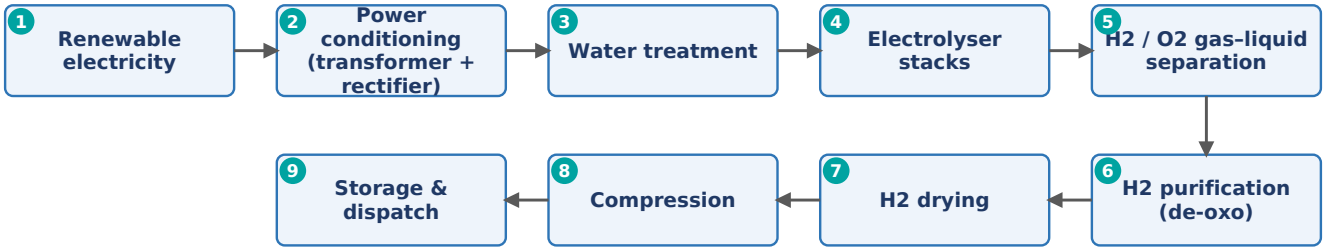
MODULE 3

Detailed Production Process

Step-by-step: green H2 by electrolysis · grey/blue H2 by SMR

3.1 Green hydrogen by water electrolysis

Process chain – each numbered block is explained below



Step 1 – Renewable electricity

Aspect	Details
Purpose	Supply low-carbon power; the largest cost item of green H_2 [R1]
Inputs	Solar / wind / hydro power, grid backup or PPA
Outputs	AC power to plant switchyard
Equipment	Solar/wind plant, HV line, substation, battery (optional)
Operating parameters	Capacity factor sets electrolyser operating hours
Process control	Energy management system, forecasting, power-purchase scheduling
Common problems	Intermittency lowers utilisation and raises cost per kg
Safety	HV electrical safety, arc-flash protection

Step 2 – Power conditioning

Aspect	Details
Purpose	Convert AC to the low-voltage, high-current DC the stacks need
Inputs	AC from grid/renewables
Outputs	Controlled DC to each stack
Equipment	Transformer, thyristor or IGBT rectifier, harmonic filters, DC busbars
Operating parameters	Stack current set-point follows power available; ripple and power factor per OEM
Process control	Current/voltage control per stack, power-factor correction
Common problems	Harmonics, rectifier losses, heat in busbars
Safety	High DC currents – insulation monitoring, lock-out/tag-out

Step 3 – Water treatment

Aspect	Details
Purpose	Produce high-purity feed water; impurities poison catalysts and membranes
Inputs	Raw water (municipal, ground, treated waste or desalinated sea water)
Outputs	Demineralised water; reject brine
Equipment	Filters, softener, reverse osmosis (RO), electro-deionisation (EDI) / mixed-bed polishers
Operating parameters	Water quality to stack OEM specification (conductivity); 18–24 kg raw water per kg H_2 incl. treatment losses [R1]
Process control	Conductivity/TOC analysers, automatic regeneration
Common problems	Membrane fouling, scaling, silica breakthrough
Safety	Chemical handling (acids, caustic, antiscalant)

Step 4 – Electrolyser stacks

Aspect	Details
Purpose	Split water into H_2 (cathode) and O_2 (anode)
Inputs	Water (or KOH solution), DC power
Outputs	Wet H_2, wet O_2, heat
Equipment	Cell stacks, end plates, manifolds, circulation pumps, heat exchangers (Module 4)
Operating parameters	AEL 70–90 °C, 1–30 bar; PEM 50–80 °C, <70 bar; SOEC 700–850 °C [R1]
Process control	Cell-voltage monitoring, temperature, pressure balance (H_2 vs O_2 side), H_2-in-O_2 analyser
Common problems	Degradation (voltage rise), gas crossover at low load, membrane thinning, catalyst loss
Safety	H_2-in-O_2 must stay well below the lower explosive limit; trip on high crossover

Step 5 – Gas–liquid separation

Aspect	Details
Purpose	Remove entrained water/electrolyte from the gas and return liquid to the stacks
Inputs	Two-phase gas/liquid from stacks
Outputs	Gas to purification; liquid back to circulation
Equipment	Separator drums (H_2 and O_2 side), demisters, level control valves
Operating parameters	Level and pressure held equal on both sides (critical for alkaline)
Process control	Level control, differential-pressure control
Common problems	Level imbalance can push gas across the separator; electrolyte carry-over
Safety	Overpressure protection, relief valves to safe vent

Step 6 – Hydrogen purification (de-oxo)

Aspect	Details
Purpose	Remove trace oxygen that crossed over
Inputs	Wet H_2 with trace O_2
Outputs	H_2 with O_2 removed (+ a little water)
Equipment	Catalytic deoxidiser (precious-metal catalyst) – O_2 reacts with H_2 to form water
Operating parameters	Inlet O_2 content, catalyst temperature
Process control	O_2 analyser upstream and downstream
Common problems	Catalyst poisoning, temperature rise if O_2 high
Safety	Exothermic reaction – high-O_2 trip

Step 7 – Hydrogen drying

Aspect	Details
Purpose	Reduce moisture to the dew point required by the user (e.g. fuel-cell grade ISO 14687)
Inputs	Saturated H_2
Outputs	Dry H_2
Equipment	Temperature-swing adsorption (TSA) dryer with twin molecular-sieve beds, coolers, knock-out pot
Operating parameters	Dew point specification per end use
Process control	Dew-point analyser, automatic bed switching and regeneration
Common problems	Bed saturation, regeneration gas losses
Safety	Hot regeneration gas is hydrogen – leak tightness

Step 8 – Compression

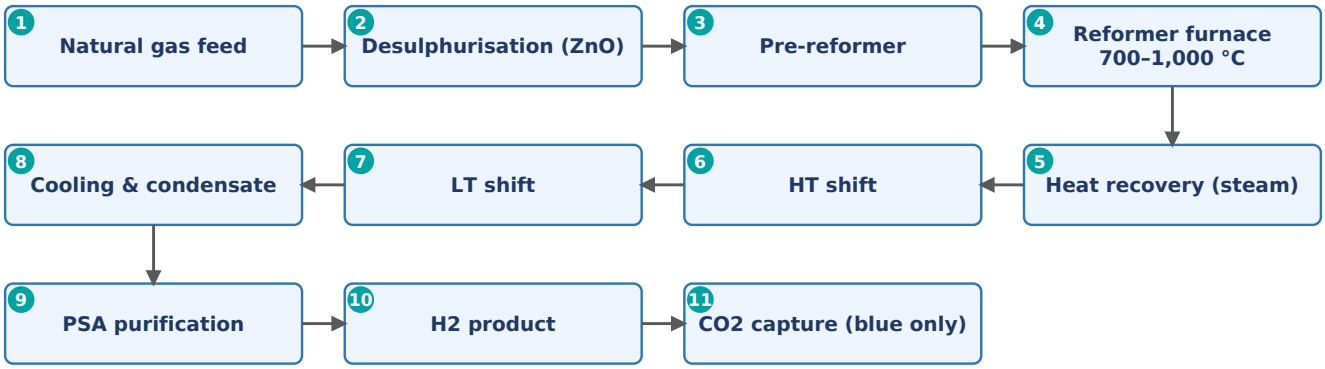
Aspect	Details
Purpose	Raise pressure for storage, pipelines or vehicles
Inputs	Dry H_2 at electrolyser outlet pressure (atmospheric – 30 bar typical [R1])
Outputs	H_2 at storage / delivery pressure
Equipment	Diaphragm, reciprocating piston or ionic compressors; intercoolers
Operating parameters	Multi-stage with intercooling; 70 bar reduces volume ~65×, 1,000 bar ~625× vs atmospheric [R1]
Process control	Suction/discharge pressure and temperature, vibration, diaphragm leak detection
Common problems	Valve and seal wear, diaphragm failure, oil contamination (lubricated machines)
Safety	High pressure + H_2; embrittlement-resistant materials; relief & ESD

Step 9 – Storage & dispatch

Aspect	Details
Purpose	Buffer production vs. demand and deliver to customers
Inputs	Compressed H_2
Outputs	H_2 to pipeline, tube trailers or end user
Equipment	Pressure vessels / cylinder banks, pipelines, dispensers, tube-trailer filling bays
Operating parameters	Storage pressure set by vessel rating (Module 8)
Process control	Inventory (pressure), flow metering, quality sampling
Common problems	Fatigue from pressure cycling
Safety	Separation distances, gas detection, fire protection

3.2 Grey / blue hydrogen by SMR – process walk-through

SMR chain (reformer conditions per US DOE [R2])



Step	Purpose	Key equipment	Watch-outs
Desulphurisation	Protect nickel catalyst from sulphur	Hydrogenator + ZnO beds	Bed breakthrough
Reforming	$CH_4 + H_2O \rightarrow CO + 3H_2$	Catalyst tubes in fired furnace	Tube metal temperature, carbon formation (keep steam/carbon ratio)
Heat recovery	Use hot syngas & flue gas to make steam	Waste-heat boiler, steam drum	Boiler water quality
Shift (HT/LT)	$CO + H_2O \rightarrow CO_2 + H_2$	Fixed-bed reactors	Catalyst activity, temperature control
PSA	Remove CO_2, CO, CH_4, H_2O	Multi-bed adsorbers, valve skid	Valve reliability, recovery vs purity
CO_2 capture	Make blue hydrogen	Amine absorber/stripper, compression	Energy for regeneration, storage site

Quiz & revision – Module 3

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Why is a de-oxo unit needed after a PEM/AEL stack?
a) To add oxygen **b)** To remove crossover O2 **c)** To remove CO2 **d)** To cool the gas

Q2. Typical raw water use per kg H2 including treatment losses (IRENA):
a) 2–4 kg **b)** 9 kg **c)** 18–24 kg **d)** 100 kg

Q3. Which dryer type uses twin molecular-sieve beds with heat regeneration?
a) PSA **b)** TSA **c)** Membrane **d)** Cyclone

Q4. Why must sulphur be removed before SMR?
a) It poisons the nickel catalyst **b)** It increases yield **c)** It is a product **d)** It reduces steam use

Q5. Compressing H2 from 1 to 70 bar reduces its volume by about:
a) 7× **b)** 65× **c)** 625× **d)** 870×

Answer key
Q1 – b: trace O2 is catalytically reacted to water; **Q2 – c**: [R1]; **Q3 – b**: temperature-swing adsorption; **Q4 – a**: ; **Q5 – b**: [R1]

MODULE 4

Electrolyser Technology Deep Dive

Cell → stack → system · AEL · PEM · SOEC · AEM

4.1 Basic concept: the electrolysis cell

Every electrolyser cell has the same five functional layers: a **positive electrode (anode)** where oxygen is formed, a **negative electrode (cathode)** where hydrogen is formed, an **ion conductor** (liquid electrolyte, polymer membrane or ceramic) between them, **porous transport layers** that let water in and gas out, and **bipolar plates** that carry current to the next cell and separate the gases. Cells are stacked in series (like batteries in a torch) to build voltage; stacks are combined in parallel to build capacity.

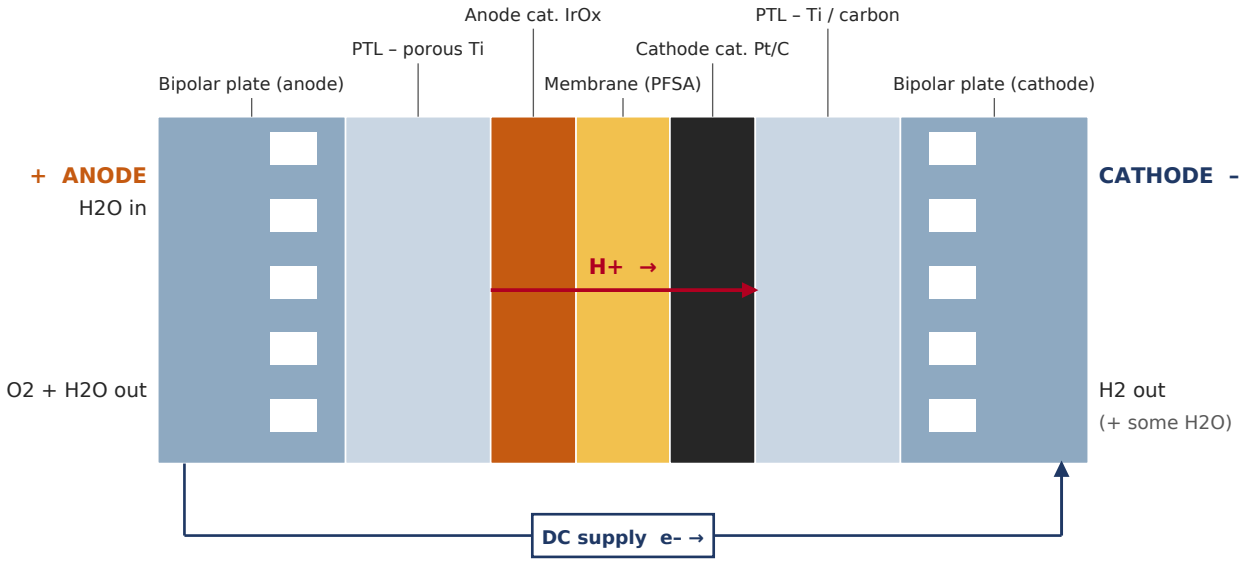


Figure 4.1 – PEM cell cross-section. Water is fed to the anode; protons (H^+) cross the membrane; electrons travel through the external DC circuit; H_2 leaves the cathode. Materials per IRENA Table 1 [R1].

4.2 Electrochemical reactions at each electrode

Type	Anode (+) – oxygen side	Cathode (-) – hydrogen side	Ion carried
Alkaline (AEL)	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$	$4H_2O + 4e^- \rightarrow 2H_2 + 4OH^-$	OH^- (through porous diaphragm)
PEM	$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$	$4H^+ + 4e^- \rightarrow 2H_2$	H^+ (through PFSA membrane)
AEM	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$	$4H_2O + 4e^- \rightarrow 2H_2 + 4OH^-$	OH^- (through polymer membrane)
SOEC	$2O^{2-} \rightarrow O_2 + 4e^-$	$2H_2O + 4e^- \rightarrow 2H_2 + 2O^{2-}$	O^{2-} (through YSZ ceramic)

Engineering principle – voltage, current and kilograms

- **Reversible voltage** 1.23 V (25 °C) = minimum electrical work (ΔG). **Thermoneutral voltage** 1.48 V = total energy (ΔH, HHV). Real cells run above 1.48 V; the excess becomes heat that the cooling system removes.
- **Faraday's law:** hydrogen output ∝ current. 2 electrons per H_2 molecule → 1 kg H_2 needs ≈ 26.6 kAh of charge (2 × 96,485 C/mol ÷ 2.016 g/mol).
- **Energy per kg** ≈ 26.6 kAh × cell voltage. At 1.8 V/cell: ≈ 47.9 kWh/kg at stack level (before rectifier and balance-of-plant). This is why lowering cell voltage is the main efficiency lever.
- **Current density** (A/cm²) sets how much H_2 each m² of cell makes – higher density = smaller, cheaper stack but higher voltage.

4.3 Stack construction – exploded view

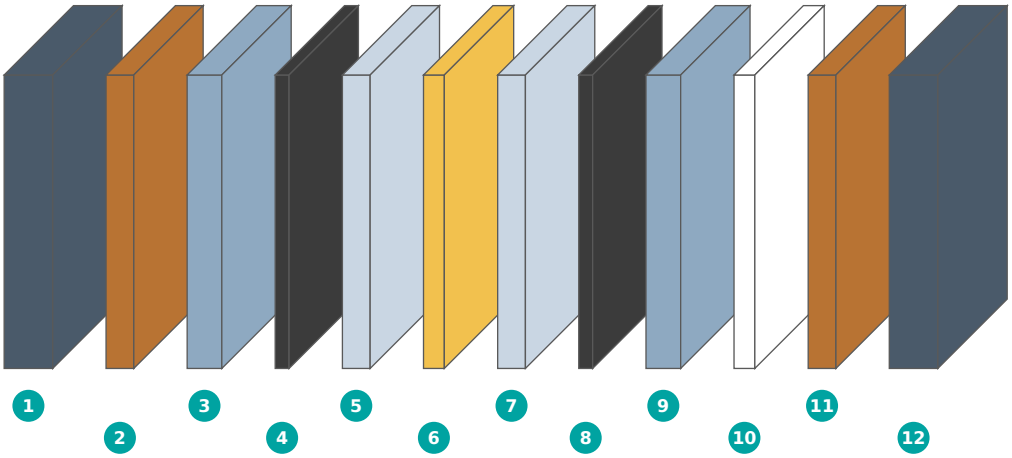


Figure 4.2 – Exploded view of a filter-press stack (numbers match the table).

#	Component	Function	Typical material (IRENA Table 1 [R1])
1	End plate	Clamps the stack; distributes compression	Steel
2	Current collector	Brings DC current into the stack	Copper / nickel-plated
3	Bipolar plate	Conducts current between cells; separates gases; flow channels	AEL/AEM: Ni-coated stainless; PEM: Pt/Au-coated Ti; SOEC: Co-coated stainless (cathode)
4	Seal / gasket	Prevents internal & external leakage	AEL: PSU, PTFE, EPDM; PEM: PTFE, PSU, ETFE; AEM: PTFE, silicone; SOEC: ceramic glass
5	Porous transport layer (anode)	Feeds water, removes O_2, conducts current	AEL: Ni mesh; PEM: Pt-coated sintered Ti; AEM: Ni foam; SOEC: Ni mesh/foam
6	Electrode + membrane (CCM)	Reaction site + ion conductor	PEM: IrO_x anode, Pt/C cathode, PFSA; AEL: Ni-coated steel + ZrO_2/PPS diaphragm; SOEC: LSCF/LSM, YSZ, Ni/YSZ
7	PTL (cathode)	Removes H_2, conducts current	PEM: sintered Ti or carbon cloth; AEM: Ni foam / carbon cloth
8–9	Seal + next bipolar plate	Starts the next cell in series	As above
11–12	Collector + end plate + tie rods	Close the stack; maintain clamping force	Steel tie rods, disc springs

4.4 Balance of plant (system level)

Sub-system	What it does	Key points
Power supply & rectifier	AC→DC at stack voltage/current	Efficiency and harmonics affect total kWh/kg

Sub-system	What it does	Key points
Gas–liquid separation	Separators on H_2 and O_2 sides	Level/pressure balance; alkaline lye return
Water circulation	Feeds water/lye, removes heat & gas	Pumps, filters, ion-exchanger (PEM keeps water ultra-pure)
Cooling system	Removes heat from overvoltage losses	Plate heat exchangers + cooling tower/dry coolers
Gas purification	De-oxo + dryer	Delivers spec purity and dew point
Control system	PLC/DCS, cell-voltage monitoring, SIS	Trips on H_2-in-O_2, pressure, temperature, low flow

4.5 Technology-by-technology

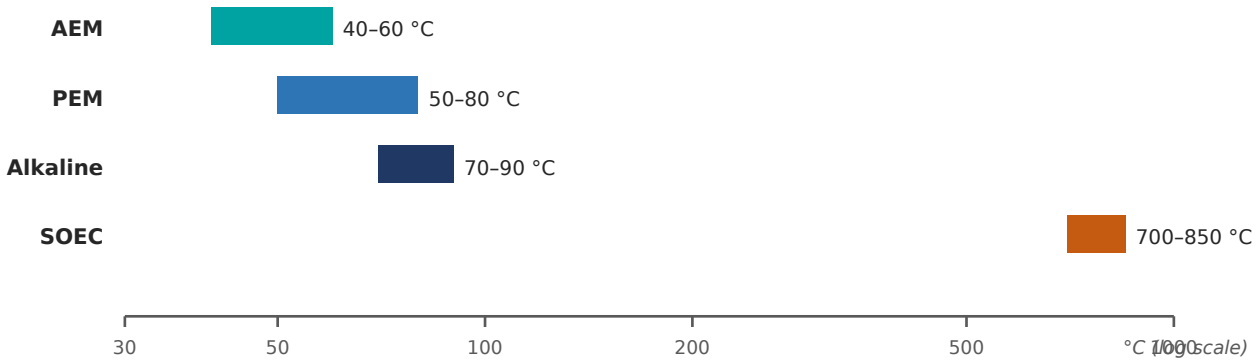
Alkaline (AEL). Simple stack, easy to manufacture; electrode areas up to ~3 m²; concentrated KOH electrolyte; robust ZrO_2-based diaphragms; nickel-coated stainless electrodes. Gas intermixing through the porous diaphragm limits low-load operation and high pressure; modern zero-gap designs have narrowed the gap to PEM. Classic designs have reached lifetimes above 30 years [R1].

PEM. Thin (~0.2 mm) PFSA membrane with advanced electrodes gives higher efficiency and current density, compact footprint and fast dynamic response; it can produce hydrogen at higher pressure. Needs precious metals (iridium, platinum) and titanium components [R1].

AEM. Alkaline chemistry with a solid polymer membrane – aims to combine AEL's cheap materials with PEM's compact design; membrane mechanical stability and durability are still being developed (IRENA classed it lab-scale in 2020) [R1].

SOEC. Ceramic cells at 700–850 °C split steam. Part of the energy can be supplied as heat (e.g. industrial waste heat), giving the highest electrical efficiency; durability and scale are the challenges (lab-scale in IRENA 2020) [R1].

Operating temperature window by electrolyser type (IRENA, 2020)



4.6 Comprehensive comparison (IRENA, 2020)

Parameter	Alkaline	PEM	AEM	SOEC
Operating temperature	70–90 °C	50–80 °C	40–60 °C	700–850 °C
Operating pressure	1–30 bar	<70 bar	<35 bar	1 bar
Electrolyte	KOH 5–7 mol/L	PFSA membrane	DVB polymer + KOH or $NaHCO_3$	Yttria-stabilised zirconia
Separator	ZrO_2 + PPS mesh	Membrane itself	Membrane itself	Ceramic itself
O_2-side catalyst	Ni-coated perforated SS	Iridium oxide	Ni or NiFeCo alloys	Perovskite (LSCF, LSM)
H_2-side catalyst	Ni-coated perforated SS	Pt nanoparticles on carbon	High-surface Ni	Ni/YSZ
System energy, 2020	50–78 kWh/kg	50–83 kWh/kg	57–69 kWh/kg	45–55 kWh/kg

Parameter	Alkaline	PEM	AEM	SOEC
Lifetime, 2020	60,000 h	50,000–80,000 h	>5,000 h	<20,000 h
Stack capex >1 MW, 2020	USD 270/kW	USD 400/kW	–	>USD 2,000/kW
System capex >10 MW, 2020	USD 500–1,000/kW	USD 700–1,400/kW	–	–
2050 targets	<45 kWh/kg; 100,000 h	<45 kWh/kg; 100–120 kh	<45 kWh/kg; 100 kh	<40 kWh/kg; 80 kh
Status (IRENA 2020)	Commercial	Commercial	Lab-scale	Lab-scale

Source: IRENA (2020), *Green Hydrogen Cost Reduction*, Table 1 and Table ES1 [R1]. Both tables report 2020 status – check current OEM data sheets before design work.

Quiz & revision – Module 4

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Which ion moves through a PEM membrane?
a) OH– **b)** H+ **c)** O2– **d)** K+

Q2. Thermoneutral voltage at 25 °C:
a) 1.23 V **b)** 1.48 V **c)** 2.0 V **d)** 0.9 V

Q3. Charge needed per kg H2 (Faraday):
a) ≈2.66 kAh **b)** ≈26.6 kAh **c)** ≈266 kAh **d)** ≈39.4 kAh

Q4. Which electrolyser uses yttria-stabilised zirconia?
a) AEL **b)** PEM **c)** AEM **d)** SOEC

Q5. PEM anode catalyst per IRENA:
a) Nickel **b)** Iridium oxide **c)** Platinum on carbon **d)** Perovskite

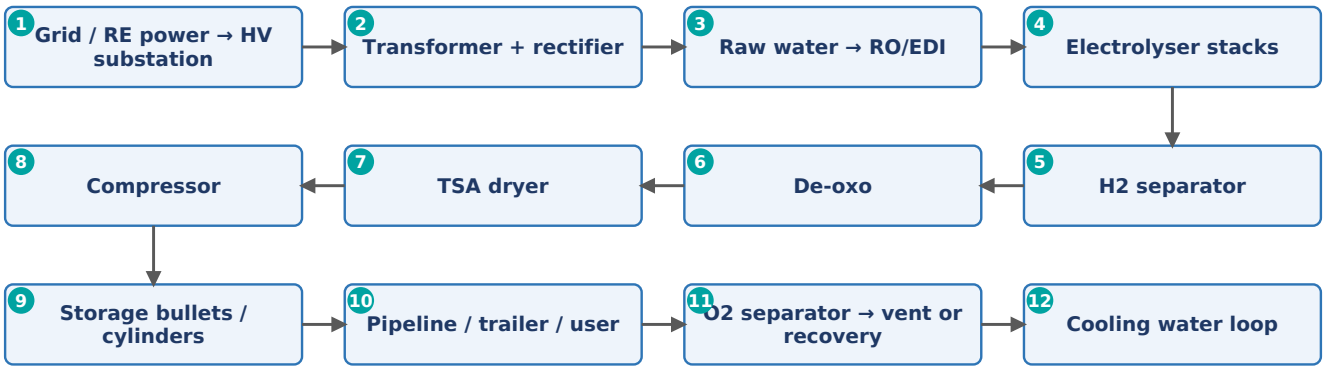
Answer key
Q1 – b: protons; **Q2 – b**: ΔH/(2F); **Q3 – b**: 2F per mol H2; **Q4 – d**: ceramic O2– conductor; **Q5 – b**: IrOx

MODULE 5

Industrial Plant Layout & Equipment

PFD · layout · utilities · instrumentation · equipment explorer

5.1 Process flow diagram (green hydrogen plant)



Blocks 11 and 12 run in parallel with the main H_2 path.

5.2 Typical plant layout

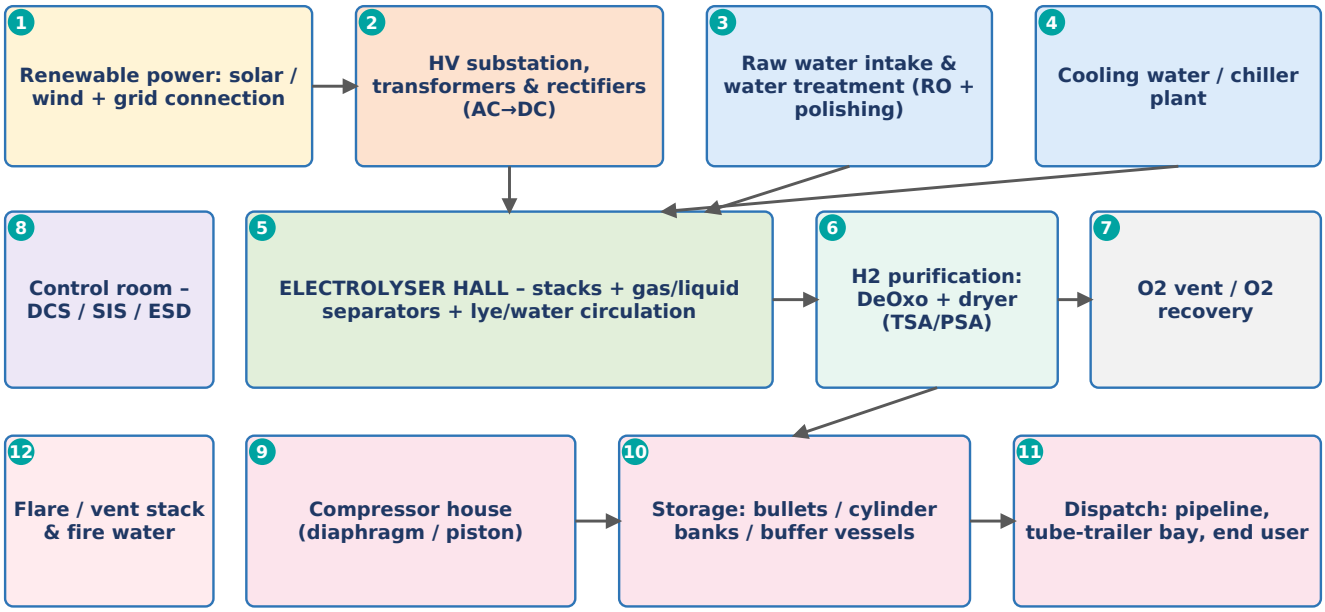


Figure 5.1 – Indicative block layout (not to scale). Hazardous gas areas (5, 6, 9, 10, 11) are separated from the control room (8) and electrical buildings (2); actual separation distances come from the site hazard study and the applicable codes (Module 10).

5.3 Utilities and infrastructure

Utility	Role	Design notes
Electrical	Main power for stacks + BoP	HV substation, transformers, rectifiers, MCCs, UPS for control, earthing & lightning protection
Water	Feed for electrolysis	Raw water storage, RO + EDI/polishing, demin tank; plan for 18–24 kg/kg H_2 [R1]
Cooling	Removes stack & compressor heat	Closed loop with plate HX; cooling tower or dry coolers

Utility	Role	Design notes
Nitrogen	Purging & inerting	Required for start-up, shutdown, maintenance purges
Instrument air	Valve actuation	Dry, oil-free
Fire water	Fire protection & cooling of vessels	Hydrants, deluge on storage per hazard study

5.4 Instrumentation & control

- **DCS / PLC** – production control: stack current, temperatures, pressures, levels, flow.
- **Safety Instrumented System (SIS)** – independent trips: H_2-in-O_2 high, pressure high/low, separator level, gas detection, ESD push-buttons.
- **Analysers** – O_2 in H_2, H_2 in O_2, dew point, water conductivity.
- **Fire & gas system** – H_2 detectors at high points, UV/IR flame detectors (H_2 flames are hard to see).

5.5 Equipment explorer

Each card gives: working principle · main components · plant function · operating parameters · maintenance points.

Equipment: Transformer-rectifier	Details
Working principle	Steps down voltage and converts AC to DC with thyristor or IGBT bridges
Main components	Transformer, rectifier bridge, filters, DC busbars
Function in plant	Supplies controlled DC to each stack
Operating parameters	Output current/voltage per stack rating
Maintenance	Thermal imaging of connections; cooling fans/water; harmonic checks

Equipment: Reverse osmosis + EDI	Details
Working principle	Pressure forces water through semi-permeable membranes; EDI polishes with ion-exchange + electric field
Main components	Pre-filters, high-pressure pump, RO vessels, EDI modules
Function in plant	Makes demin water
Operating parameters	Conductivity at outlet to OEM spec
Maintenance	Membrane cleaning (CIP), antiscalant dosing, filter changes

Equipment: Electrolyser stack	Details
Working principle	Electrochemical water splitting (Module 4)
Main components	Cells, bipolar plates, seals, end plates
Function in plant	Produces H_2 and O_2
Operating parameters	Temperature, pressure, current density per type [R1]
Maintenance	Cell-voltage trending (degradation), re-torque, stack refurbishment

Equipment: Gas-liquid separator	Details
Working principle	Gravity + demister separate liquid from gas
Main components	Vessel, demister pad, level transmitters
Function in plant	Dries gas, returns liquid
Operating parameters	Level & pressure balanced H_2/O_2 sides
Maintenance	Level instrument calibration, demister inspection

Equipment: De-oxo reactor	Details
Working principle	Catalytic recombination of O_2 with H_2
Main components	Catalyst bed, heater, O_2 analyser
Function in plant	Removes O_2
Operating parameters	Bed temperature; inlet O_2
Maintenance	Catalyst replacement; analyser calibration

Equipment: TSA dryer	Details
Working principle	Adsorbs water on molecular sieve; regenerated by heating
Main components	Twin beds, heater, cooler, switching valves
Function in plant	Dries H_2
Operating parameters	Outlet dew point
Maintenance	Valve leak tests; sieve replacement

Equipment: PSA unit (SMR plants)	Details
Working principle	Impurities adsorb at high pressure, desorb at low pressure
Main components	Multiple adsorber beds, valve skid, tail-gas drum
Function in plant	Purifies H_2
Operating parameters	Cycle time; recovery vs purity
Maintenance	Valve maintenance; adsorbent life

Equipment: Diaphragm compressor	Details
Working principle	Hydraulically driven metal diaphragm compresses gas – no contact with oil
Main components	Diaphragm set, hydraulic system, valves
Function in plant	Raises pressure, oil-free
Operating parameters	Stages & intercooling per pressure ratio
Maintenance	Diaphragm leak detection & replacement

Equipment: Storage vessel / cylinder bank	Details
Working principle	Holds compressed H_2
Main components	Shell, nozzles, PRVs, instruments
Function in plant	Buffer storage
Operating parameters	Design pressure per code
Maintenance	Periodic inspection, fatigue assessment, PRV testing

Equipment: Gas detector	Details
Working principle	Catalytic, thermal-conductivity or electrochemical sensing
Main components	Sensor head, transmitter
Function in plant	Leak detection → alarm/trip
Operating parameters	Set-points as % of LFL per hazard study
Maintenance	Bump test & calibration

Quiz & revision – Module 5

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Which system trips the plant on high H2-in-O2?
a) DCS **b)** SIS **c)** SCADA historian **d)** Water analyser

Q2. Why are diaphragm compressors common for H2?
a) Cheapest **b)** Oil-free gas path **c)** No cooling needed **d)** Single stage to 900 bar

Q3. Nitrogen in an H2 plant is mainly used for:
a) Cooling **b)** Purging/inerting **c)** Electrolyte **d)** Fuel

Q4. Where should H2 detectors be located indoors?
a) Near floor **b)** At high points / ceiling **c)** Outside only **d)** In drains

Q5. Which unit is typical of SMR plants rather than electrolysis plants?
a) De-oxo **b)** Rectifier **c)** PSA **d)** EDI

Answer key
Q1 – b: independent safety layer; **Q2 – b**: ; **Q3 – b**: ; **Q4 – b**: H2 is buoyant; **Q5 – c**: removes CO2/CO/CH4

MODULE 6

Process Simulation (Calculator)

Worksheet · worked example · sensitivities

Read this first – simplified estimate, not plant data
This module is a **pencil-and-calculator simulator**. It uses physics (stoichiometry, Faraday's law, ideal compression) plus a few published reference values (IRENA 2020). Items marked **[assumption]** are teaching inputs chosen for illustration – they are **not** vendor guarantees or real project costs. Replace them with OEM data and actual tariffs for any real study.

6.1 Inputs you can change

Input	Symbol	Example value	Basis
Electrolyser capacity	P (MW)	10 MW	[assumption] – your choice
Specific energy consumption	SEC (kWh/kg)	51.2 kWh/kg (= 65 % LHV)	IRENA 2020 reference case [R1]
Operating hours (renewable availability)	h (h/yr)	4,000 h/yr	[assumption] e.g. solar + wind hybrid
Electricity price	p (USD/MWh)	30 USD/MWh	[assumption]; IRENA uses 20 USD/MWh in sensitivities [R1]
System capex	C (USD/kW)	750 USD/kW	Mid-range of IRENA 2020 AEL 500–1,000 [R1]
Discount rate / life	r, n	8 % / 20 yr	8 % per IRENA [R1]; life [assumption]
Fixed O&M	–	2 % of capex per yr	[assumption]
Water use / cost	–	20 kg/kg; 2 USD/m³	IRENA 18–24 kg/kg [R1]; price [assumption]
Delivery pressure	–	30 → 350 bar	[assumption] for compression example

6.2 Equations

H_2 per hour (kg/h) = P × 1,000 ÷ SEC

H_2 per day = kg/h × 24 (at full load) · per year = kg/h × h

Electricity (MWh/yr) = P × h

O_2 (kg) = 8 × H_2 (kg) · Stoichiometric water = 9 × H_2
from 2H2O → 2H2 + O2 (36 : 4 : 32)

Efficiency (LHV) = 33.3 ÷ SEC · Efficiency (HHV) = 39.4 ÷ SEC

$CRF = r(1+r)^n \div [(1+r)^n - 1]$
capital recovery factor

LCOH = (C·P·1000·CRF + O&M) ÷ annual kg + SEC × p ÷ 1,000 + water
levelised cost of hydrogen, USD/kg

Ideal isothermal compression work = $(RT/M)\cdot\ln(P_2/P_1)$
divide by a realistic compressor efficiency

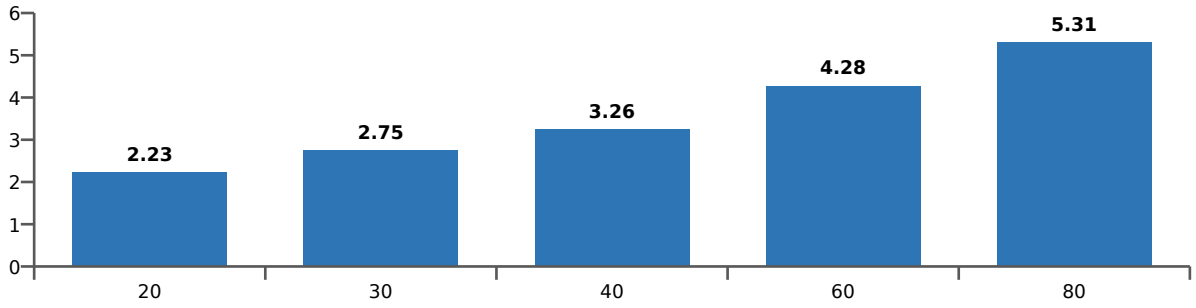
6.3 Worked example (10 MW, 4,000 h/yr)

Output	Calculation	Result
H_2 per hour	10 × 1,000 ÷ 51.2	195 kg/h
H_2 per day (full load)	195 × 24	4.69 t/day
H_2 per year	195 × 4,000	781 t/yr
Electricity	10 MW × 4,000 h	40,000 MWh/yr
Oxygen by-product	8 × H_2	6250 t/yr
Water (incl. treatment)	20 × H_2	15625 m³/yr
Efficiency	33.3 ÷ 51.2 (LHV) · 39.4 ÷ 51.2 (HHV)	65 % LHV · 77 % HHV
Capex share of LCOH	750×10,000×CRF ÷ annual kg	0.98 USD/kg
Fixed O&M	2 % × capex ÷ annual kg	0.19 USD/kg
Electricity	51.2 × 30 ÷ 1,000	1.54 USD/kg
Water	0.020 m³ × 2 USD	0.04 USD/kg
LCOH (simplified)	sum	**2.75 USD/kg**
Compression 30→350 bar (ideal)	(RT/M)·ln(350/30)	0.84 kWh/kg (≈1.7 kWh/kg at 50 % efficiency [assumption])

6.4 Sensitivity: LCOH (USD/kg) vs. electricity price and operating hours

Hours \ Price →	20 USD/MWh	30 USD/MWh	40 USD/MWh	60 USD/MWh	80 USD/MWh
2000 h/yr	3.40	3.92	4.43	5.45	6.48
3000 h/yr	2.62	3.14	3.65	4.67	5.70
4000 h/yr	2.23	2.75	3.26	4.28	5.31
6000 h/yr	1.84	2.36	2.87	3.89	4.92
8000 h/yr	1.65	2.16	2.67	3.70	4.72

Simplified LCOH at 4,000 h/yr vs electricity price (USD/MWh)



Lesson: electricity price drives cost once hours are high; at low hours the capex share rises steeply. This matches IRENA's finding that electricity is the largest cost component and capex the second [R1].

6.5 Worksheet – your own case

Input / output	Your value	Notes
Capacity P (MW)		
SEC (kWh/kg)		
Hours (h/yr)		
Electricity price (USD/MWh)		
Capex (USD/kW)		

Input / output	Your value	Notes
H2 per hour (kg/h)		
H2 per year (t)		
Electricity (MWh/yr)		
O2 (t/yr)		
Water (m^3/yr)		
Efficiency LHV / HHV		
LCOH (USD/kg)		

Simplified vs real performance

- Real SEC rises with stack **degradation** and at **part load** (balance-of-plant loads stay fixed).
- Stack replacement (IRENA reference: 80,000 h stack life [R1]) adds a mid-life cost not modelled here.
- Purity (de-oxo + drying) and compression add electricity; storage and transport add cost (Module 8).
- Taxes, land, grid charges, financing structure and subsidies (e.g. India SIGHT incentives) are excluded.

Quiz & revision – Module 6

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. 10 MW at 50 kWh/kg makes how much H2 per hour?
a) 20 kg **b)** 200 kg **c)** 2,000 kg **d)** 500 kg

Q2. Oxygen produced per kg H2:
a) 1 kg **b)** 2 kg **c)** 8 kg **d)** 16 kg

Q3. SEC of 51.2 kWh/kg equals what LHV efficiency?
a) ~45 % **b)** ~65 % **c)** ~77 % **d)** ~90 %

Q4. At low operating hours, which LCOH component grows most?
a) Electricity **b)** Capex share **c)** Water **d)** Oxygen credit

Q5. Largest cost component of green H2 (IRENA):
a) Water **b)** Electricity **c)** Land **d)** Compression

Answer key

Q1 – b: 10,000 ÷ 50; **Q2 – c**: 32 ÷ 4; **Q3 – b**: 33.3 ÷ 51.2; **Q4 – b**: fixed cost spread over fewer kg; **Q5 – b**: [R1]

MODULE 7

Purification & Conditioning

Separation · drying · PSA · membranes · de-oxo · compression · liquefaction

7.1 Why purify?

Raw hydrogen from any process carries impurities: water and traces of O_2 (electrolysis), or CO_2, CO, CH_4, N_2 and H_2O (reforming/gasification). The end use sets the spec – for example, PEM fuel cells are very sensitive to CO and sulphur, so vehicle hydrogen must meet **ISO 14687**, while a refinery hydrotreater tolerates more impurities.

Technology	Principle	Removes	Where used
Gas–liquid separation	Gravity + demister	Liquid water / electrolyte droplets	Directly after every electrolyser
Drying – TSA	Adsorption on molecular sieve; heat regeneration	Moisture	Electrolysis plants; pipeline gas
Drying – refrigeration	Cool to condense water	Bulk moisture (pre-step)	Ahead of TSA / compressors
PSA	Impurities adsorb at high pressure, release at low pressure	CO_2, CO, CH_4, N_2, H_2O	SMR/ATR/gasification; by-product H_2 recovery
Membrane separation	H_2 permeates faster through polymer/metal membrane	Bulk separation; Pd membranes give very high purity	Refinery off-gas recovery, small units
Deoxidation (de-oxo)	Catalytic $2H_2 + O_2 \rightarrow 2H_2O$	Trace O_2	Electrolysis plants
Cryogenic separation	Partial condensation at low temperature	Hydrocarbons, N_2	Large refinery / petrochemical streams

7.2 Pressure Swing Adsorption – how a cycle works



Several beds run out of phase so product flow is continuous. Hydrogen is barely adsorbed, so it passes through; the tail gas (impurities + some H_2) is burned in the SMR furnace. Trade-off: higher purity ↔ lower hydrogen recovery.

7.3 Compression

- **Reciprocating piston** – high flow and pressure; needs careful sealing (lubricated or oil-free designs).
- **Diaphragm** – metal diaphragm isolates gas from oil; standard for high-purity, high-pressure duty such as refuelling stations.
- **Ionic liquid** – liquid piston; few moving parts.
- **Electrochemical** – proton-conducting cell pumps H_2; no moving parts, small scale.
- Electrolysers can deliver pressurised gas directly (up to ~30 bar typical, higher possible), but higher stack pressure increases gas crossover and losses [R1].

7.4 Liquefaction

Hydrogen is cooled below –253 °C in stages (pre-cooling with liquid nitrogen or mixed refrigerant, then a hydrogen Claude/Brayton cycle with expansion turbines). An **ortho-to-para conversion** catalyst is required because the natural ortho→para conversion releases heat that would otherwise boil off stored liquid. Liquefaction reduces volume ~870× vs atmospheric gas [R1] but consumes a large share of the hydrogen's energy content, so it is used mainly for long-distance or high-volume delivery.

7.5 Storage

Compressed, liquid, underground, hydride and chemical-carrier storage are covered in Module 8.

Quiz & revision – Module 7

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Which step removes trace oxygen from electrolytic H2?
a) PSA **b)** De-oxo **c)** Membrane **d)** Cyclone

Q2. In PSA, what happens to impurities?
a) Pass through **b)** Adsorb at high pressure **c)** React to water **d)** Freeze

Q3. Why is ortho–para conversion needed in liquefaction?
a) To increase density **b)** Conversion heat would boil off stored LH2 **c)** To remove CO **d)** To raise pressure

Q4. Fuel-cell vehicle hydrogen quality standard:
a) ISO 9001 **b)** ISO 14687 **c)** IEC 61850 **d)** ASME VIII

Q5. Main drawback of higher PSA purity:
a) Lower H2 recovery **b)** More CO2 **c)** Higher water use **d)** No drawback

Answer key
Q1 – b: ; **Q2 – b**: then released at low pressure; **Q3 – b**: ; **Q4 – b**: ; **Q5 – a**:

MODULE

8

Storage & Transportation

Compressed · liquid · underground · hydrides · pipelines · trailers · ships

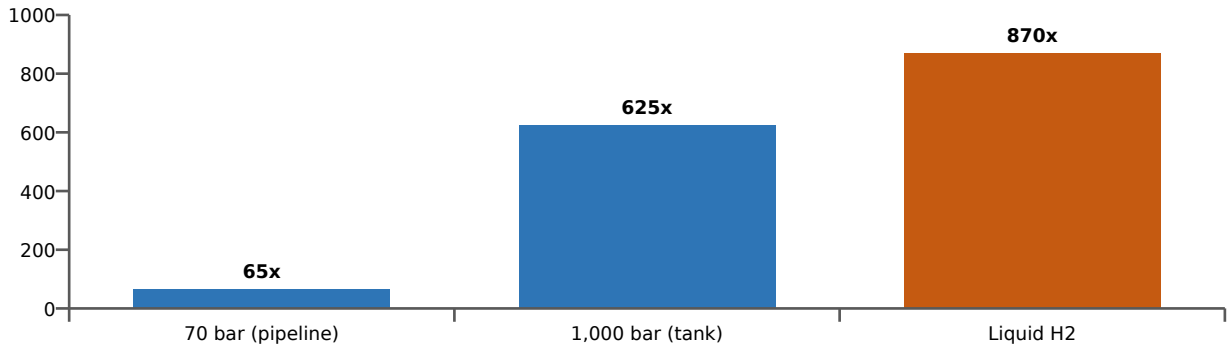
8.1 Storage methods

Method	How it works	Typical use	Advantages	Limitations
Compressed gas	Steel or composite vessels; vehicle tanks at 350 or 700 bar	Vehicles, refuelling stations, plant buffers	Simple, fast fill/discharge	Low volumetric density; compression energy
Liquid H_2	Cryogenic, vacuum-insulated tanks at –253 °C	Bulk delivery, space, shipping	High density per tank	Liquefaction energy; boil-off
Underground (salt cavern)	Gas stored in solution-mined caverns	Seasonal / large-scale storage	Very large volume at low cost per kg	Geology-dependent; few sites
Metal hydrides	H_2 absorbed into metal alloy lattice	Small stationary, submarines, niche	Low pressure, compact, safe	Heavy; heat management
Chemical carriers (NH_3, LOHC, methanol)	H_2 bound in a liquid; released by cracking/dehydrogenation	Export / long distance	Liquid handling with existing assets	Conversion & reconversion losses

8.2 Pressure vessel types

Type	Construction	Notes
Type I	All-metal (steel)	Heaviest, cheapest; stationary
Type II	Metal liner, hoop-wrapped composite	Stationary
Type III	Metal (aluminium) liner, fully wrapped composite	Lighter; transport
Type IV	Polymer liner, fully wrapped carbon fibre	Lightest; 700 bar car tanks

Volume reduction vs. atmospheric-pressure H2 (x times smaller)



Source: IRENA (2020) citing BNEF (2019) [R1].

8.3 Transportation

Mode	Description	Best for
Pipelines	Dedicated H_2 lines (design code ASME B31.12) or repurposed gas lines after integrity assessment	Large, steady flows over short–medium distance
Tube trailers	Bundles of high-pressure cylinders on a truck	Small–medium customers, refuelling stations
Liquid H_2 tankers	Cryogenic road tankers	Larger volumes over longer road distances

Mode	Description	Best for
Ships	LH_2 carriers (first demonstration ship operated Australia → Japan) or ammonia/LOHC tankers	Intercontinental trade
On-site production	Electrolyser at the point of use	Avoids transport altogether

Engineering challenge – embrittlement in transport
High-strength steels can lose ductility in hydrogen (Module 10). Pipeline repurposing therefore needs material testing, lower design stress and fatigue checks under pressure cycling.

Quiz & revision – Module 8

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Vehicle tank pressures commonly used:
a) 35 & 70 bar **b)** 350 & 700 bar **c)** 1,000 & 2,000 bar **d)** 10 & 20 bar

Q2. Which tank type uses a polymer liner with full carbon-fibre wrap?
a) I **b)** II **c)** III **d)** IV

Q3. Best option for seasonal, very large storage:
a) Tube trailers **b)** Salt caverns **c)** Metal hydride **d)** Type IV tanks

Q4. Design code for hydrogen piping & pipelines:
a) ASME B31.12 **b)** API 650 **c)** IS 456 **d)** ISO 9001

Q5. Main drawback of chemical carriers like ammonia:
a) Toxic & conversion losses **b)** Too light **c)** Need 700 bar **d)** Cannot be shipped

Answer key
Q1 – b: ; **Q2 – d**: ; **Q3 – b**: ; **Q4 – a**: ; **Q5 – a**: cracking back to H2 costs energy

MODULE 9

Industrial Applications

Steel · refining · ammonia · methanol · fuel cells · transport · storage

Refineries

Hydrotreating removes sulphur from fuels; hydrocracking breaks heavy oil into lighter products.

$$R\text{–}S + 2H_2 \rightarrow R\text{–}H + H_2S \text{ (desulphurisation, simplified)}$$

Largest single user today; replacing grey with green/blue H_2 cuts refinery emissions directly.

Ammonia & fertiliser

Haber–Bosch synthesis of ammonia; ammonia makes urea and other fertilisers.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

India's green ammonia tenders under the National Green Hydrogen Mission target this use.

Methanol

Syngas or CO_2 + H_2 converted over copper-based catalyst.

$$CO + 2H_2 \rightarrow CH_3OH \cdot CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$$

Chemical feedstock and marine fuel.

Steel (DRI)

Hydrogen reduces iron ore to sponge iron in a shaft furnace, then an electric arc furnace makes steel.

$$Fe_2O_3 + 3H_2 \rightarrow 2Fe + 3H_2O$$

Replaces coke/coal in the blast furnace – water instead of CO_2.

Fuel cells

Electrochemical reverse of electrolysis: H_2 + O_2 → electricity + heat + water.

$$H_2 + \tfrac{1}{2}O_2 \rightarrow H_2O + \text{electricity}$$

PEM for vehicles; SOFC for stationary power.

Heavy-duty transport

Fuel-cell or H_2-combustion trucks and buses; ammonia/methanol for ships.

India's NGHM pilots: 70 hydrogen buses and trucks, 16 refuelling stations (see Hydrogen Vehicle Study deck).

Energy storage

Surplus renewable power → electrolysis → storage → fuel cell or turbine.

Long-duration and seasonal storage; low round-trip efficiency.

Chemical industry

Hydrogenation of oils, production of hydrogen peroxide, pharmaceuticals, electronics.

Small volumes, often high purity.

9.1 India context

India's National Green Hydrogen Mission (launched 4 January 2023, outlay ₹19,744 crore to FY 2029-30) targets about 5 MMT per year of green hydrogen production by 2030, with incentives for

electrolyser manufacturing and green hydrogen production (SIGHT) and pilots in steel, mobility and shipping [R9]. Refining and fertiliser are the obvious first users because they already consume grey hydrogen.

Quiz & revision – Module 9

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Reaction for ammonia synthesis:
a) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ **b)** $CO + 2H_2 \rightarrow CH_3OH$ **c)** $C + H_2O \rightarrow CO + H_2$ **d)** $2H_2O \rightarrow 2H_2 + O_2$

Q2. In H2-DRI steelmaking the by-product is:
a) CO_2 **b)** Water **c)** Slag only **d)** Methane

Q3. Largest current hydrogen consumer:
a) Fuel-cell cars **b)** Refining **c)** Aviation **d)** Glass

Q4. Which fuel cell type is common in vehicles?
a) SOFC **b)** PEMFC **c)** MCFC **d)** PAFC

Q5. India's NGHM 2030 production target:
a) 0.5 MMT/yr **b)** ~5 MMT/yr **c)** 50 MMT/yr **d)** 100 MMT/yr

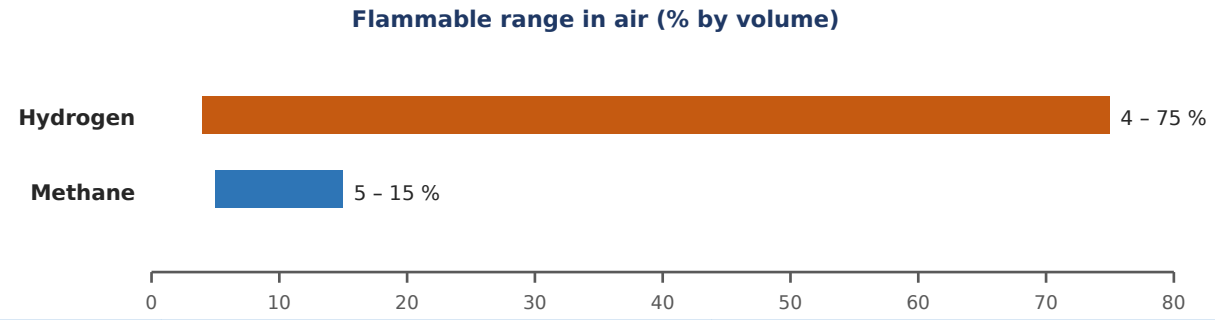
Answer key
Q1 – a: ; **Q2 – b**: ; **Q3 – b**: ; **Q4 – b**: ; **Q5 – b**:

MODULE 10

Safety & Industrial Standards

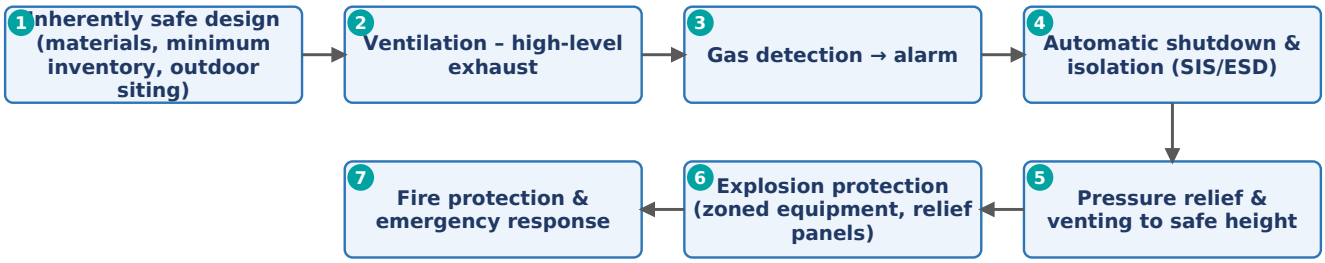
Hazards · protection layers · area classification · codes

10.1 Hazard properties



Hazard	Why it matters	Control
Flammability	Ignites from 4 % to 75 % in air; very low ignition energy (~0.017 mJ) [R6]	Leak prevention, ventilation, ignition-source control, bonding/earthing
Leakage	Smallest molecule – leaks through fittings, seals and some materials	Welded joints where possible, leak testing (helium/H_2), detectors
Invisible flame	Pale flame hard to see in daylight; low radiant heat	UV/IR flame detectors; thermal cameras
Embrittlement	Atomic H diffuses into steel and reduces ductility, especially high-strength steels and welds	Material selection (austenitic stainless, low-strength steels), hardness limits, fatigue design
Pressure	Stored energy in high-pressure vessels	Relief valves, burst discs, code-designed vessels
Cryogenic (LH_2)	Cold burns; air condensation on cold lines; boil-off pressure rise	Vacuum insulation, relief, PPE
Asphyxiation	Displaces oxygen in confined spaces	O_2 monitoring, confined-space permits

10.2 Protection layers



10.3 Hazardous area classification

Zone (IEC 60079-10-1)	Meaning	Hydrogen equipment rating
Zone 0	Explosive atmosphere present continuously or for long periods	Ex equipment for Group IIC
Zone 1	Likely in normal operation	Group IIC, temperature class T1 or better
Zone 2	Not likely; if it occurs, only briefly	Group IIC, T1 or better

Hydrogen is in gas group **IIC** (the most easily ignited group, with acetylene). Its auto-ignition temperature (≈585 °C) puts it in temperature class **T1** (equipment surface ≤ 450 °C).

10.4 Key standards and codes

Standard	Scope
ISO 22734	Hydrogen generators using water electrolysis – industrial, commercial and residential
ISO 14687	Hydrogen fuel quality – product specification
ISO 19880-1	Gaseous hydrogen – fuelling stations
ISO/TR 15916	Basic considerations for the safety of hydrogen systems
IEC 60079 series	Explosive atmospheres – area classification & Ex equipment
NFPA 2	Hydrogen Technologies Code (USA)
ASME B31.12	Hydrogen piping and pipelines
IEC 61511	Functional safety – safety instrumented systems for process industry
India	PESO approvals; Gas Cylinders Rules 2016; Static & Mobile Pressure Vessels (Unfired) Rules; MNRE Green Hydrogen Standard (2023)

Always check the latest edition and the local authority having jurisdiction.

10.5 Safety scenarios (think, then read the answer)

Scenario
A hydrogen detector in the electrolyser hall alarms at a low % of LFL during start-up.

Recommended response
Do not reset blindly. Follow the alarm response: stop ignition-source work, confirm with a portable detector, locate the leak (often new flanges/fittings), isolate and depressurise via the safe vent, purge with N_2, repair, leak-test, then restart.

Scenario
The H_2-in-O_2 analyser on an alkaline electrolyser rises at low load.

Recommended response
Classic crossover issue: at low current, gas production falls but diffusion through the diaphragm does not – the H_2 fraction in O_2 rises. Increase load or shut down per the SIS trip limit, well below the lower explosive limit.

Scenario
A contractor wants to weld a bracket on a pipe rack next to a live hydrogen line.

Recommended response
Hot work needs a permit, gas test, isolation/positive isolation of the line or a safe distance, fire watch, and continuous gas monitoring.

Scenario
A crack is found at a weld in a high-pressure hydrogen buffer vessel after many pressure cycles.

Recommended response
Likely hydrogen-assisted fatigue. Take the vessel out of service, perform NDT and fitness-for-service assessment; review material, hardness and cycle count against design.

Quiz & revision – Module 10

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Gas group for hydrogen in IEC 60079:
a) IIA **b)** IIB **c)** IIC **d)** I

Q2. Temperature class for hydrogen:
a) T1 **b)** T3 **c)** T4 **d)** T6

Q3. Where do hydrogen leaks accumulate indoors?
a) Floor pits **b)** Under roofs / high points **c)** Evenly **d)** In drains

Q4. Embrittlement is worst in:
a) Low-strength steels **b)** High-strength steels & hard welds **c)** Austenitic stainless **d)** Copper

Q5. Standard for electrolysis hydrogen generators:
a) ISO 22734 **b)** ISO 14001 **c)** IEC 61850 **d)** API 610

Answer key
Q1 – c: ; **Q2 – a**: AIT ≈ 585 °C; **Q3 – b**: buoyant; **Q4 – b**: ; **Q5 – a**:

Appendix A – Flashcards

Cover the right column and test yourself, or print and cut along the lines.

Energy carrier	A medium that stores/transports energy produced elsewhere (H2, electricity).
LHV of H2	120 MJ/kg = 33.3 kWh/kg (water as vapour).
HHV of H2	142 MJ/kg = 39.4 kWh/kg (water as liquid).
SMR reaction	CH4 + H2O → CO + 3H2 (endothermic).
Water-gas shift	CO + H2O → CO2 + H2 (exothermic).
POX reaction	CH4 + ½O2 → CO + 2H2 (exothermic).
Methane pyrolysis	CH4 → C(s) + 2H2; no CO2 formed.
Reversible cell voltage	1.23 V at 25 °C (ΔG).
Thermoneutral voltage	1.48 V at 25 °C (ΔH, HHV).
Charge per kg H2	≈ 26.6 kAh (Faraday's law, 2 e– per H2).
PEM anode reaction	2H2O → O2 + 4H+ + 4e–.
Alkaline cathode reaction	4H2O + 4e– → 2H2 + 4OH–.
SOEC electrolyte	Yttria-stabilised zirconia (YSZ), conducts O2–.
PEM catalysts	Anode IrOx; cathode Pt on carbon.
Alkaline electrolyte	KOH solution, 5–7 mol/L (IRENA).
Bipolar plate	Conducts current cell-to-cell, separates gases, carries flow channels.
PTL	Porous transport layer – water in, gas out, current through.
CCM	Catalyst-coated membrane – the PEM reaction heart.
PSA	Pressure swing adsorption – impurities adsorb at high P, desorb at low P.
TSA	Temperature swing adsorption dryer – molecular sieve regenerated by heat.
De-oxo	Catalytic removal of O2 by reaction with H2 to water.
SEC	Specific energy consumption, kWh per kg H2.
LCOH	Levelised cost of hydrogen, USD (or ₹) per kg.
Crossover	Gas permeating to the wrong electrode side – efficiency and safety issue.
Embrittlement	Loss of ductility of metals exposed to hydrogen.
Gas group IIC	Most easily ignited gas group (H2, acetylene).

Temperature class T1	Max equipment surface temp 450 °C – H2 rating.
Type IV tank	Polymer liner + full carbon-fibre wrap; 700 bar vehicles.
Salt cavern	Large-scale underground H2 storage.
Ortho–para conversion	Catalysed during liquefaction to avoid boil-off heat.
ISO 14687	Hydrogen fuel quality specification.
ISO 22734	Standard for water-electrolysis hydrogen generators.
DRI	Direct reduced iron – H2 reduces ore: $Fe_2O_3 + 3H_2 \rightarrow 2Fe + 3H_2O$.
Haber–Bosch	$N_2 + 3H_2 \rightleftharpoons 2NH_3$ – ammonia synthesis.

Appendix B – Knowledge Assessment

B1. Process sequencing

Put these green-hydrogen steps in order: (a) TSA drying (b) rectifier (c) de-oxo (d) electrolyser stack (e) compression (f) water treatment (g) gas–liquid separator (h) storage.

Put these SMR steps in order: (a) PSA (b) reformer (c) desulphurisation (d) LT shift (e) HT shift (f) heat recovery.

B2. Equipment identification

Description	Your answer
Vessel with demister that returns lye to the stack	
Twin molecular-sieve beds regenerated by heat	
Metal plate with flow channels that separates two cells	
Device converting AC to DC for the stack	
Catalyst bed that turns trace O2 into water	
Compressor with metal membrane isolating gas from oil	
Ceramic oxide-ion conductor operating near 800 °C	
Multi-bed adsorber unit burning its tail gas in the furnace	

B3. Chemical reactions – complete and balance

- $CH_4 + H_2O \rightarrow$ ____ + ____
- $CO + H_2O \rightarrow$ ____ + ____
- PEM anode: $2H_2O \rightarrow$ ____ + ____ + ____
- Fe_2O_3 + ___$H_2 \rightarrow$ ___Fe + ___H_2O
- N_2 + ___$H_2 \rightleftharpoons$ ___NH_3

B4. Calculation questions

- A 5 MW PEM plant runs at 55 kWh/kg. How many kg H_2 per hour, and how many tonnes of O_2 per day at full load?
- A stack operates at 1.9 V/cell. Estimate stack-level energy per kg H_2.
- Electricity costs 40 USD/MWh and SEC is 52 kWh/kg. What is the electricity cost per kg?

B5. Industrial case studies

Refinery switch to green H2
A refinery uses 30 t/day of grey hydrogen from SMR. Management wants to replace 20 % with green hydrogen. List the plant units needed, estimate the electrolyser size at 52 kWh/kg running 24 h, and name two integration risks.

Alkaline vs PEM selection
A solar-only site (no storage) will feed an electrolyser. Which technology would you shortlist and why? Consider dynamic response, minimum load, footprint, materials and cost (use Module 4 table).

Refuelling station supply
A bus depot needs fuel-cell-grade hydrogen at 350 bar. Choose between on-site electrolysis and tube-trailer delivery; list purification, compression and safety requirements.

Answer key

- B1 green: f → b → d → g → c → a → e → h. SMR: c → b → f → e → d → a.
- B2: separator; TSA dryer; bipolar plate; transformer-rectifier; de-oxo; diaphragm compressor; SOEC (YSZ) cell; PSA.
- B3: $CO + 3H_2$; $CO_2 + H_2$; $O_2 + 4H^+ + 4e^-$; 3, 2, 3; 3, 2.
- B4: 5,000 ÷ 55 = 90.9 kg/h; $O_2 = 8 \times 90.9 \times 24 \div 1{,}000 = 17.5$ t/day. 26.6 kAh × 1.9 V ≈ 50.5 kWh/kg. 52 × 40 ÷ 1,000 = 2.08 USD/kg.
- B5 (refinery): 6 t/day = 250 kg/h × 52 = 13 MW electrolyser (24 h) + water treatment, purification, compression to the hydrogen header pressure, buffer storage; risks: intermittent supply vs steady refinery demand, purity/pressure matching, power-supply reliability.
- B5 (solar site): PEM usually shortlisted for fast dynamics and wide load range; alkaline for lower capex if a battery/grid smooths supply.
- B5 (bus depot): ISO 14687 quality, drying + de-oxo (electrolysis) or quality testing (trailer), multi-stage compression to above 350 bar for cascade storage, ISO 19880-1 station design, gas & flame detection, area classification.

Appendix C – Glossary (alphabetical, searchable in any PDF reader with Ctrl+F)

Term	Meaning
AEL	Alkaline electrolysis
AEM	Anion exchange membrane electrolysis
ATR	Autothermal reforming
BoP	Balance of plant – everything except the stack
CCS	Carbon capture and storage
CCM	Catalyst-coated membrane
CRF	Capital recovery factor
DCS	Distributed control system
DRI	Direct reduced iron
EDI	Electro-deionisation
ESD	Emergency shutdown
HHV / LHV	Higher / lower heating value
HRS	Hydrogen refuelling station
LCOH	Levelised cost of hydrogen
LFL / LEL	Lower flammable / explosive limit
LH2	Liquid hydrogen
LOHC	Liquid organic hydrogen carrier
NGHM	India's National Green Hydrogen Mission
PEM	Proton exchange (polymer electrolyte) membrane
PFSA	Perfluorosulfonic acid membrane (e.g. Nafion)
POX	Partial oxidation
PSA	Pressure swing adsorption
PTL	Porous transport layer
RO	Reverse osmosis
SEC	Specific energy consumption (kWh/kg)
SIS	Safety instrumented system
SMR	Steam methane reforming
SOEC	Solid oxide electrolysis cell
TSA	Temperature swing adsorption
WGS	Water-gas shift
YSZ	Yttria-stabilised zirconia

Appendix D – Learning Progress Tracker

Tick when completed, write your quiz score (out of 5), and list topics to revise.

Module	Done ☐	Date	Quiz score /5	Topics mastered	Topics to revise
1 Introduction	☐				
2 Production technologies	☐				
3 Detailed process	☐				
4 Electrolyser deep dive	☐				
5 Plant layout & equipment	☐				
6 Process simulation	☐				
7 Purification & conditioning	☐				
8 Storage & transport	☐				
9 Applications	☐				
10 Safety & standards	☐				
Appendix B assessment	☐				

Mastery rule
5/5 = mastered · 4/5 = revise flashcards for the missed item · 3/5 or less = re-read the module and redo the quiz after 2 days.

Appendix E – References

Ref	Source	Link
[R1]	IRENA (2020). Green Hydrogen Cost Reduction: Scaling up Electrolysers to Meet the 1.5 °C Climate Goal. Table 1, Table ES1, pp. on water use and compression.	https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2020/Dec/IRENA_Green_hydrogen_cost_2020.pdf
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[R3]	IEA (2024). Global Hydrogen Review 2024 – Hydrogen demand.	https://www.iea.org/reports/global-hydrogen-review-2024/hydrogen-demand
[R4]	IEA (2025). Global Hydrogen Review 2025 – Demand.	https://www.iea.org/reports/global-hydrogen-review-2025/demand
[R5]	IEA (2023). Towards Hydrogen Definitions Based on their Emissions Intensity – Executive summary.	https://www.iea.org/reports/towards-hydrogen-definitions-based-on-their-emissions-intensity/executive-summary
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[R7]	NREL (2019). Manufacturing Cost Analysis for PEM and Alkaline Water Electrolyzers (technology characteristics).	https://www.nrel.gov/docs/fy19osti/70380.pdf
[R8]	Renewable Watch (2025). Global Hydrogen Review 2025: IEA (demand ~100 Mt in 2024).	https://renewablewatch.in/2025/09/21/global-hydrogen-review-2025-iea/
[R9]	MNRE / PIB. National Green Hydrogen Mission; pilot projects for hydrogen buses and trucks (3 Mar 2025).	https://nghm.mnre.gov.in/admin/uploads/174110589023911Press%20Release_PIB_H2_fuelled_buses_trucks.pdf
[R10]	MNRE (2023). Green Hydrogen Standard for India (≤2 kg CO2-eq/kg H2, well-to-gate).	https://mnre.gov.in/

Standard reaction enthalpies, Faraday constant, cell voltages and ideal-gas compression are textbook physical chemistry values. Process descriptions without a numeric reference are general engineering knowledge and are intentionally kept qualitative.